INCH-POUND

MIL-C-60315B (AR)
9 September 1991
SUPERSEDING
MIL-C-60315A (AR)
2 February 1973

MILITARY SPECIFICATION

# CHARGE, PROPELLING, M67, FOR 105MM HOWITZER LOADING, ASSEMBLING AND PACKING

This specification is approved for use by the US Army Armament, Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers the loading, assembling and packing for one type of propelling charge designated as Charge, Propelling, M67, for Howitzer, M2A2, M103 and M137.

## 2. APPLICABLE DOCUMENTS

2.1 Government documents

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

PPP-C-843 - Cushioning Material, Cellulosic

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1315

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MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for

STANDARDS

MILITARY

MIL-STD-1168 - Lot Numbering of Ammunition

(Unless otherwise indicated, copies of the federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg 4D, 700 Robbins Ave., Philadephia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

US ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

9205472 - Charge, Propelling, M67 Assembly
9288695 - Charge, Propelling, M67 Assembly
9345265 - Container, Steel
9381476 - Packing and Marking for Containers, Steel for Bulk Propellants

(Copies of drawings, publications and other Government documents required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with applicable drawings and specifications.



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3.2 Propelling charge assembly. The propelling charge assembly shall comply with all requirements specified on Drawing (dwg.) 9205472 or 9288695 and with all requirements specified in applicable specifications.

3.3 Increment loading assembly weight. The net weight of the propellant shall not differ by more than the amounts specified in Table I from the loading authorization:

TABLE I
Weight Tolerances for Propelling Charge Increments

| Increment Number | Tolerance (Ounces) |
|---|---|
| 1 | 0.02 |
| 2 | 0.015 |
| 3 | 0.015 |
| 4 | 0.015 |
| 5 | 0.015 |
| 6 | 0.02 |
| 7 | 0.02 |

3.4 Charge, propelling assembly weight. The charge, propelling assembly weight shall not differ by more than 0.10 ounces from the total of the amount specified in the loading authorization, plus tare weight of bags and cord (see 6.3).

3.5 Workmanship. All parts and assemblies shall be fabricated and finished in a thorough, workmanlike manner. All parts shall be free of dirt, grease, and other foreign material. All required marking shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's



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overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as Quality Conformance Inspection (see 4.3).

4.3 Quality conformance inspection.

4.3.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078 and MIL-STD-1168. In addition, inspection lots of parts for charges shall contain:

a. Single perforated propellant from not more than one lot.

b. Multi perforated propellant from not more than one lot.

4.3.2 Examination and tests. See MIL-A-48078.

a. Classification of characteristics. Quality conformance examinations and tests are specified in the following Classification of Characteristics paragraphs. The contractor's quality program or detailed inspection system shall provide assurance of compliance of all characteristics with the applicable drawings and specification requirements utilizing as a minimum the conformance criteria specified. When cited herein, attributes sampling inspection shall be conducted in accordance with Table II below, using the inspection levels stated in the Classification of Characteristics paragraphs.



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Table II
Attributes sampling inspection.

| Lot Size | Major Sample Size | Minor Sample Size |
|---|---|---|
| 2 to 8 | * | See note |
| 9 to 15 | * | 5 |
| 16 to 25 | * | 5 |
| 26 to 50 | 32 | 5 |
| 51 to 90 | 32 | 12 |
| 91 to 150 | 32 | 12 |
| 151 to 280 | 32 | 14 |
| 281 to 500 | 32 | 17 |
| 501 to 1200 | 74 | 20 |
| 1201 to 3200 | 74 | 24 |
| 3201 to 10000 | 87 | 30 |
| 10001 to 35000 | 109 | 36 |
| 35001 to 150000 | 124 | 40 |
| 150001 to 500000 | 156 | 40 |
| 500001 and over | 187 | 40 |

* - 100 % Inspection

NOTE: Use 100% inspection when lot size is less than 5 units. Use sample size when lot is greater than 5 units.

b. Alternative quality conformace provisions. Alternative quality conformance procedures, methods, or equipment, such as statistical process control, tool control, other types of sampling procedures, etc. may be used by the contractor when they provide, as a minimum the level of quality assurance required by the provisions specified herein. Prior to applying such alternative procedures, methods, or equipment, the contractor shall describe them in a written proposal submitted to the Government for evaluation (see 6.6). When required, the contractor shall demonstrate that the effectiveness of each proposed alternative is equal to or better than the specified quality assurance provision(s) herein. In cases of dispute as to whether the contractor's proposed alternative(s) provides equivalent assurance, the provisions of this specification shall apply. All approved alternative provisions shall be specifically incorporated into the contractor's quality program or detailed inspection system, as applicable.



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QUALITY CONFORMANCE INSPECTION

# CLASSIFICATION OF CHARACTERISTICS

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| PARAGRAPH | TITLE | SHEET | DRAWING NUMBER |
|---|---|---|---|
| 4.3.2.1 | Bag increment | 1 OF 1 | 9205470 or 9288697 |
| | | | NEXT HIGHER ASSEMBLY: 9205471 or 9288696 |

| CLASSIFICATION | EXAMINATION OR TEST | CONFORMANCE CRITERIA | REQUIREMENT PARAGRAPH | INSPECTION METHOD REFERENCE |
|---|---|---|---|---|
| Critical | None defined. | | | |
| Major | | | | |
| 101 | Foil missing or improperly assembled (applicable to increment 5 only). | Table II | 3.2 | Visual |
| Minor | | | | |
| 201 | Any seam or opening incompletely stiched. | Table II | 3.2 | Visual |
| 202 | Marking misleading or unidentifiable. | Table II | 3.2 | Visual |
| 203 | Evidence of poor workmanship. | Table II | 3.2 | Visual |

NOTES:





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# CLASSIFICATION OF CHARACTERISTICS



| PARAGRAPH | TITLE | SHEET | DRAWING NUMBER |
|---|---|---|---|
| 4.3.2.2 | Increment loading assembly | 1 OF 1 | 9205471 or 9288696 |

NEXT HIGHER ASSEMBLY: 9205472 or 9288695

| CLASSIFICATION | EXAMINATION OR TEST | CONFORMANCE CRITERIA | REQUIREMENT PARAGRAPH | INSPECTION METHOD REFERENCE |
|---|---|---|---|---|
| Critical | None defined. | | | |
| Major | | | | |
| 101 | Assembly damaged to the extent that propellant can escape. | Table II | 3.2 | Visual |
| Minor | | | | |
| 201 | Any seam broken or incompletely stitched | Table II | 3.2 | Visual |
| 202 | Thread broken. | Table II | 3.2 | Visual |
| 203 | Evidence of poor workmanship. | Table II | 3.5 | Visual |

NOTES:



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# CLASSIFICATION OF CHARACTERISTICS



| PARAGRAPH | TITLE | SHEET | DRAWING NUMBER |
|---|---|---|---|
| 4.3.2.3 | Charge, Assembly | 1 OF 1 | 9205472 or 9288695 |
| | | | NEXT HIGHER ASSEMBLY |

| CLASSIFICATION | EXAMINATION OR TEST | CONFORMANCE CRITERIA | REQUIREMENT PARAGRAPH | INSPECTION METHOD REFERENCE |
|---|---|---|---|---|
| Critical | | | | |
| 1 | Missing or incorrect increments. | 100 % 1/ | 3.2 | Scale |
| Major | | | | |
| 101 | Incorrect sequence of increments. | 100 % | 3.2 | Visual |
| 102 | Assembly damaged to the extent that propellant can escape. | 100 % | 3.2 | Visual |
| Minor | | | | |
| 201 | Increments insecurely tied at junctions. | Table II | 3.2 | Visual |
| 202 | Ties loose, broken or insecurely tied. | Table II | 3.2 | Visual |
| 203 | Marking missing or unidentifiable. | Table II | 3.2 | Visual |
| 204 | Evidence of poor workmanship. | Table II | 3.5 | Visual |

NOTES:

1/ The defect shall be classed critical when the weight is in excess of 3 times that specified in 3.4. Major when the weight is in excess of that specified in 3.4 but not greater than 3 times.





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QUALITY CONFORMANCE INSPECTION

# CLASSIFICATION OF CHARACTERISTICS



| PARAGRAPH | TITLE | SHEET | DRAWING NUMBER |
|---|---|---|---|
| 4.3.2.4 | Drum, prior to filling | 1 OF 1 | 9345265 |
| | | | NEXT HIGHER ASSEMBLY: 9381476 |

| CLASSIFICATION | EXAMINATION OR TEST | CONFORMANCE CRITERIA | REQUIREMENT PARAGRAPH | INSPECTION METHOD REFERENCE |
|---|---|---|---|---|
| Critical | None defined. | | | |
| Major | | | | |
| 101 | Foreign material in drum. | 100 % | 3.2 | Visual |
| 102 | Inspection of gasket. | 100 % | 3.2 | Visual |
| Minor | None defined. | | | |

NOTES:



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# CLASSIFICATION OF CHARACTERISTICS



| PARAGRAPH | TITLE | SHEET 1 OF 1 | DRAWING NUMBER |
|---|---|---|---|
| 4.3.2.5 | Drum sealed | | 9345265 and 9381476 |
| | | | NEXT HIGHER ASSEMBLY |

| CLASSIFICATION | EXAMINATION OR TEST | CONFORMANCE CRITERIA | REQUIREMENT PARAGRAPH | INSPECTION METHOD REFERENCE |
|---|---|---|---|---|
| Critical | None defined. | | | |
| Major | | | | |
| 101 | Locking ring improperly positioned. | 100 % | 3.2 | Visual |
| Minor | | | | |
| 201 | Car seal missing or improperly positioned. | Table II | 3.2 | Visual |
| 202 | Marking misleading or unidentifiable. | Table II | 3.2 | Visual |

NOTES:





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4.3.3 Testing.

4.3.3.1 Increment loading assembly weight. The increment shall be weighed and check weighed 100 percent. Any increment which fails to comply with the applicable requirements shall be classed defective and removed from the lot. Inspection equipment will be in accordance with 4.3.5.

4.3.3.1.1 Method 1. The propellant shall be weighed in a container and placed in the bag, the bag sewn and the increment loading assembly weighed. The net weight of propellant may be calculated using an average tare weight of the empty bags or the actual tare weight as marked on the individual bag.

4.3.3.1.2 Method 2. The propellant in a container shall be weighed and then check weighed on a different balance and shall then be placed in the bag and the bag sewn. The weight of the loaded bags should be checked as a major defect in accordance with the sampling plan outlined in Table II. The net weight of the propellant shall be calculated using an average weight of the empty bag or the actual tare weight as marked on the individual bag.

4.3.3.2 Air pressure test (see 5.1.2). The containers shall be checked 100 % as specified in 4.4.1. If any container fails to meet the requirement, it shall be classed defective and removed from the lot.

4.3.3.3 Charge, propelling, assembly weight (see 3.4). The defects shall be classified as follows: Critical defect: Weight deviation in excess of 3 times of that specified in 3.4. Major defect: Weight deviation in excess of that specified in 3.4 but not greater than 3 times that specified in 3.4. The assemblies shall be weighed 100 % and compliance with 3.4 shall be determined by using average weight of the empty bag assembly. If the charge is outside the established weight tolerance remove increment 5 by raising it off the scale without breaking the cord and replace it with a weight representing the average weight of increment 5. If the weight deviation still does not meet the requirement 3.4, the charge shall be classed defective and removed from the lot. If the weight deviation requirement is met and if it can be determined that the deviation is caused by the lead foil (tare weight of the bag) the charge shall be considered acceptable.

4.3.4 Gasket inspection. Container lids will be inspected 100% for cut, dried-out, cracked, abraded or improperly affixed gaskets or excessive adhesive used in affixing gaskets. If any gasket is found with any of the conditions listed above, the lid shall classed defective and removed from the lot.



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4.3.5 Inspection equipment. The inspection equipment required to perform the examination and tests prescribed herein is described in the applicable paragraph. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See section 6 of MIL-A-48078 and 6.4 herein.

4.4 Methods of inspection.

4.4.1 Air pressure test. The containers shall be subjected to an internal pressure test of 2.0 - 0.5 pounds per square inch (psi) by a method satisfactory to the contracting officer's representative. A gage (designed to read 1/4 psi) or a water manometer shall be assembled in the system. A container experiencing a drop of 1/4 psi on the gage or 0.7 inches on the manometer in 15 seconds shall be classed defective and removed from the lot.

5. PACKAGING

5.1 Packing.

5.1.1 Level A. The propelling charges shall be packed in containers conforming to 9345265. The gross weight of the pack shall not exceed 150 pounds. Cushioning material conforming to PPP-C-843 shall be added as required to provide a tight pack.

5.1.1.1 Reworked containers. In addition to meeting the requirements of 9345265, the gasket of all reworked containers shall be replaced in accordance with 9345265 prior to reuse.

5.1.2 Air pressure test of containers. Immediately after packing, all containers shall be tested with compressed air at an internal air pressure of 2.0 - 0.5 psi. The container shall withstand the specified pressure for a minimum of 15 seconds without apparent leakage as defined in 4.4.1.

5.2 Marking of steel drums. Containers shall be marked in accordance with 9381476. The Proper Shipping Name and Identification Number (PSN/IN) shall be "CHARGES, PROPELLING, FOR CANNON UN0242".

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The M67 propelling charge covered in this specification is intended for use in 105MM ammunition for Howitzer, M2A2, M103, M119 and M137.



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6.2 Acquisition requirements. Acquisition documents must specify the following (see MIL-A-48078):

a. Title, number, and date of the specification.

b. Issue of the DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

6.3 Charge, propelling assembly weight. This requirement applies to the load plant only due to change in tare weight of the bags because of variation in bag material and moisture content.

6.4 Submission of designs for approval. See 6.2.3 of MIL-A-48078. Submit equipment designs, as required to Commander, ARDEC, ATTN: AMSMC-QAR-R (D), Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List in the Contract, DD Form 1423 in the contract.

6.5 Submission of ammunition data cards. In addition to the normal distribution of records, when charges are procured by the Army, one copy of all ammunition data cards shall be forwarded to each of the following:

Commander
ARDEC
ATTN: AMSMC-QAR-R (D)
SMCAR-AEE-BP
Picatinny Arsenal, NJ 07806-5000

6.6 Submission of alternative quality conformance provisions. Unless otherwise specified in the contract, proposed alternative quality conformance provisions will be submitted by the contractor for evaluation by the technical activity responsible for preparation of this specification.

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.



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6.8 Subject term (key word) listing.

Semi-fixed ammunition
Artillery ammunition
Single base propellant
Solid propellant

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1315-AD10)



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

| I RECOMMEND A CHANGE: | 1. DOCUMENT NUMBER<br>MIL-C-60315B (AR) | 2. DOCUMENT DATE (YYMMDD)<br>9 September 1991 |
|---|---|---|

3. DOCUMENT TITLE
CHARGE,PROPELLING, M67, FOR 105MM HOWITZER LAP

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

| a. NAME (*Last, First, Middle Initial*) | b. ORGANIZATION | |
|---|---|---|
| c. ADDRESS (*Include Zip Code*) | d. TELEPHONE (Include Area Code)<br>(1) Commercial<br>(2) AUTOVON (*If applicable*) | 7. DATE SUBMITTED (*YYMMDD*) |

8. PREPARING ACTIVITY

| a. NAME<br>US ARMY ARDEC<br>STANDARDIZATION OFFICE | b. TELEPHONE (*Include Area Code*)<br>(1) Commercial (201) 724-6675<br>(2) AUTOVON 880-6675 |
|---|---|
| c. ADDRESS (*Include Zip Code*)<br>ATTN: SMCAR-BAC-S<br>PICATINNY ARSENAL, NJ 07806-5000 | IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:<br>Defense Quality and Standardization Office<br>5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466<br>Telephone (703) 756-2340 AUTOVON 289-2340 |

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